Machine Learning in Finance & Insurance

Project 3: Deep Hedging

Learning trading strategies to hedge option payoffs

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November 4, 2026

Project 3 in one picture

Problem 1: Black–Scholes

Learn a hedge and compare it with analytical delta hedging.

- Constant stock volatility.
- Minimize squared hedging error.
- Compare with a known benchmark.

Problem 2: Heston

Manage residual risk with stochastic volatility and AVaR.

- Random stock variance.
- Minimize the risk of large losses.
- Compare two risk levels.

Simulate paths $\longrightarrow$ **Train a hedge** $\longrightarrow$ **Evaluate on test paths**

A European call option gives the buyer a choice

A call option gives its buyer the right, but not the obligation, to buy one share for a fixed price at a future date.

Symbol	Meaning
S_t	Stock price at time t
K	Strike: agreed purchase price
T	Maturity: exercise date
p	Premium paid at time 0

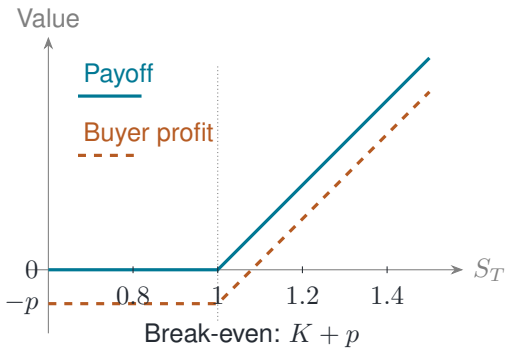
"European" means exercise only at T.

Example: $K = 1$**,** $T = 3$ **months**

- If $S_T = 1.20$: buy for 1, obtaining a share worth 1.20. Value: 0.20.
- If $S_T = 0.80$: let the option expire. Value: 0.

Call payoff: $g(S_T) = (S_T - K)^+ = \max(S_T - K, 0)$

The option payoff and the buyer's profit differ



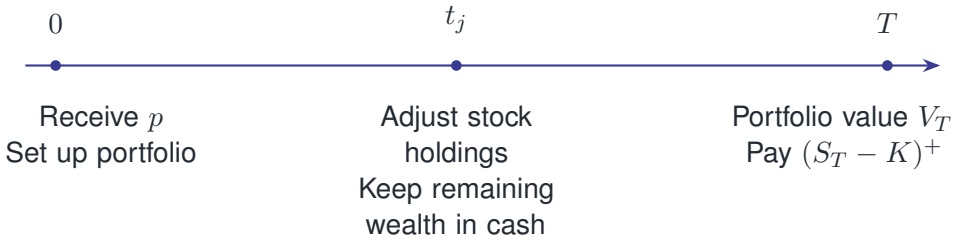
At maturity, with zero interest:

$$\text{Payoff} = (S_T - K)^+,$$
$$\text{Buyer profit} = (S_T - K)^+ - p.$$

For $K = 1$ and an *illustrative* premium $p = 0.08$:

S_T	Payoff	Profit
0.80	0	-0.08
1.00	0	-0.08
1.20	0.20	0.12

The seller uses trading to cover the final payment

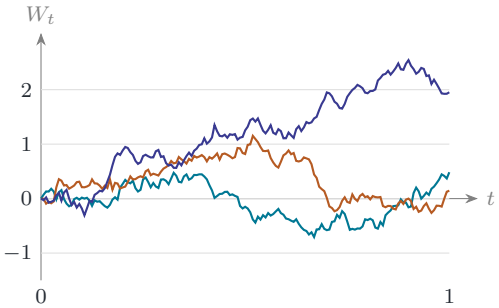


$$\boxed{\text{Seller's hedging P\&L} = V_T - (S_T - K)^+}$$

- A positive P&L is money left over; a negative P&L is a shortfall.

Goal: choose a trading strategy that makes this hedging error small.

Brownian motion: a model for continuous random movement



Three simulated paths

A standard Brownian motion W_t:

- Starts at $W_0 = 0$.
- Has continuous, irregular paths.
- Has independent increments over disjoint time intervals.

$$\boxed{W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)}$$

The increment has mean 0 and standard deviation $\sqrt{\Delta t}$.

How to generate a path on a time grid

Draw independent $Z_j \sim \mathcal{N}(0,1)$ and set $W_{t_{j+1}} = W_{t_j} + \sqrt{\Delta t}\, Z_j$.
Different draws produce different paths. Brownian motion itself is not a stock price.

Stock-price dynamics in the Black-Scholes model

In the Black-Scholes model, the stock price follows the following dynamics under the risk-neutral measure $\mathbb{Q}$:

$$\boxed{dS_t = \underbrace{rS_t\,dt}_{\text{drift}} + \underbrace{\sigma S_t\,dW_t}_{\text{random shock}}, \qquad S_0 = s_0 > 0.} \tag{1}$$

What each symbol means

S_t	Stock price at time t
r	Risk-free interest rate
σ	Volatility of returns
dW_t	Brownian increment

How to read the equation

- $rS_t dt$: expected local change.
- $\sigma S_t dW_t$: unpredictable change
- Both scale with the current price S_t.

For a short interval, the relative price change is approximately

$$\frac{\Delta S_t}{S_t} \approx r\Delta t + \sigma\sqrt{\Delta t}\,Z, \qquad Z \sim \mathcal{N}(0,1).$$

Why ordinary calculus is not enough

Suppose we want the change in $f(W_t)$. A Taylor expansion gives

$$\Delta f \approx f'(W_t)\Delta W_t + \tfrac{1}{2} f''(W_t)(\Delta W_t)^2.$$

For a smooth path $x(t)$

Δx has size Δt, $\quad (\Delta x)^2$ has size $(\Delta t)^2$.

The squared term vanishes when accumulated over a fine time grid.

This gives $df(x_t) = f'(x_t)\, dx_t$.

For Brownian motion W_t

ΔW has size $\sqrt{\Delta t}$, $\quad (\Delta W)^2$ has size Δt.

The squared term survives: over finer grids,

$$\sum_j (\Delta W_j)^2 \longrightarrow T.$$

A quick check: $f(w) = w^2$

The ordinary chain rule would give $d(W_t^2) = 2W_t\, dW_t$.

Itô's rule adds the missing term: $\boxed{d(W_t^2) = 2W_t\, dW_t + dt}$.

Itô's lemma

Let W_t be a standard Brownian motion, and let X_t satisfy

$$dX_t = a(t, X_t)\,dt + b(t, X_t)\,dW_t.$$

For a smooth function $f(t, x)$, Itô's formula is

$$\boxed{df(t, X_t) = \left(\frac{\partial f}{\partial t} + a\frac{\partial f}{\partial x} + \frac{1}{2}b^2\frac{\partial^2 f}{\partial x^2}\right)dt + b\frac{\partial f}{\partial x}\,dW_t.}$$

Special case: a function of Brownian motion itself

Set $X_t = W_t$, so $a = 0$ and $b = 1$. For $f(t, W_t)$:

$$\boxed{df(t, W_t) = \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial w^2}\right)dt + \frac{\partial f}{\partial w}\,dW_t.}$$

Here the derivatives of $f(t, w)$ are evaluated at (t, W_t).

Black–Scholes price

1. Solving the stock SDE using Itô's formula

Under the risk-neutral measure $\mathbb{Q}$, starting from $S_t = s$,

$$S_T = s \exp\left[\left(r - \frac{1}{2}\sigma^2\right)\tau + \sigma\sqrt{\tau}\, Z\right], \qquad Z \sim \mathcal{N}(0,1), \quad \tau = T - t.$$

This describes the possible stock prices at maturity.

2. Replication and no arbitrage determine today's price

Average the call payoff under $\mathbb{Q}$, then discount to today:

$$\boxed{C(t,s) = e^{-r\tau}\, \mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+ \mid S_t = s\right].}$$

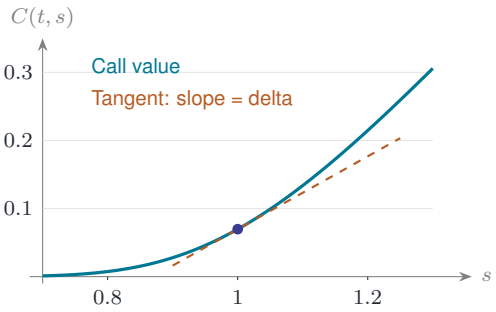
3. Calculating this expectation gives the pricing formula

Substitute the expression for S_T and evaluate the normal integral:

$$\boxed{C(t,s) = s\Phi(d_+) - Ke^{-r\tau}\Phi(d_-)}$$

$$d_+ = \frac{\log(s/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_- = d_+ - \sigma\sqrt{\tau}.$$

Delta tells us how many shares to hold



Delta is the slope of the call-price curve:

$$\boxed{H_t^{\mathrm{BS}}(s) = \frac{\partial C}{\partial s}(t,s).}$$

For a small stock-price change,

$$\Delta C \approx H_t^{\mathrm{BS}}(s)\,\Delta s.$$

If delta is 0.535, a 0.01 rise in the stock increases the option value by about 0.00535.

For one call sold, hold 0.535 shares to offset that increase in liability.

Delta hedging: recompute the holding at each trading date and rebalance using the cash account.

Minimize the terminal hedging error

We sell an option for p and trade the stock to cover its payoff $g(S_T)$.

$$\text{Hedging P\&L} = p + \int_0^T H_u \, dS_u - g(S_T).$$

Choose the trading strategy to minimize the expected squared error:

$$\inf_{H \in \mathcal{H}} \mathbb{E}^{\mathbb{Q}} \left[\left(p + \int_0^T H_u \, dS_u - g(S_T) \right)^2 \right].$$

- H_u: number of shares held at time u; $\int H_u \, dS_u$: trading gains.
- $p = C(0, s_0)$ is fixed, not learned.
- $\mathcal{H}$: predictable strategies with well-defined, square-integrable gains.

The continuous-time delta hedge achieves zero error. We will learn a hedge numerically.

Discretize the trading strategy

Use a uniform grid $t_j = jT/N$, $j = 0, \ldots, N$. Hold $H_j(S_{t_j}; \theta_j)$ shares through $[t_j, t_{j+1})$, where each $H_j : \mathbb{R} \to \mathbb{R}$ is a neural network.

$$\int_0^T H_u \, dS_u \;\approx\; \sum_{j=0}^{N-1} H_j(S_{t_j}; \theta_j)(S_{t_{j+1}} - S_{t_j}).$$

For m simulated paths, replace the expectation by an average and train jointly:

$$\min_{\theta_0, \ldots, \theta_{N-1}} \frac{1}{m} \sum_{i=1}^{m} \left[p + \sum_{j=0}^{N-1} H_j\left(s_{t_j}^{(i)}; \theta_j\right) \left(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}\right) - g\left(s_T^{(i)}\right) \right]^2 . \tag{2}$$

- i indexes paths; j indexes trading dates; θ_j are network parameters.
- Each network observes the current price and outputs the shares to hold.
- All networks learn from one terminal loss.

Problem 1(a): Verify the stock-price solution with Itô

Task: Show that geometric Brownian motion solves equation (1) with $r = 0$:

$$S_t = s_0 e^{\sigma W_t - \frac{1}{2}\sigma^2 t} \qquad \Longrightarrow \qquad dS_t = \sigma S_t \, dW_t.$$

1. Define $f(t, w) = s_0 e^{\sigma w - \frac{1}{2}\sigma^2 t}$, so $S_t = f(t, W_t)$.
2. Compute the derivatives and substitute into Itô's formula.

Problem 1(b): Simulate training and test paths

Task: Generate 100,000 training paths and 10,000 independent test paths.

$$N = 30, \qquad s_0 = 1, \qquad T = \tfrac{1}{4} \text{ year}, \qquad \sigma = 0.35, \qquad \Delta t = T/N.$$

1. Set the initial price of every path to $s_0 = 1$.
2. Draw independent $Z_{j+1} \sim \mathcal{N}(0, 1)$ for each path and time step.
3. Apply the exact update for $j = 0, \ldots, 29$:

$$\boxed{S_{t_{j+1}} = S_{t_j} \exp\left(-\tfrac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t}\, Z_{j+1}\right).}$$

Keep the time indexing consistent

30 trading intervals require 31 prices per path, including S_0 and S_T.
Training array: $100{,}000 \times 31$; test array: $10{,}000 \times 31$.

Problem 1(c): Build the neural-network hedge

Task: Implement one neural network $H_j : \mathbb{R} \to \mathbb{R}$ per trading date.

$$\underbrace{S_{t_j}}_{\text{current price}} \longrightarrow \underbrace{H_j(S_{t_j}; \theta_j)}_{\text{number of shares to hold}}$$

- Create $H_0, \ldots, H_{29}$, each with its own parameters θ_j.
- **Input:** stock price.
- **Output:** stock holding

Problem 1(d): Train by minimizing equation (2)

Task: Hedge a call with $K = 1$, $T = 1/4$, and payoff $g(S_T) = (S_T - K)^+$.

1. Compute and fix $p = C(0, s_0)$ using the Black–Scholes formula (3).
2. On each training path, form the terminal hedging error:

$$\varepsilon_i = p + \sum_{j=0}^{29} H_j(s_{t_j}^{(i)}; \theta_j)(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}) - (s_T^{(i)} - K)^+.$$

3. Jointly optimize all network parameters using the empirical loss:

$$\boxed{\mathcal{L}(\theta_0, \ldots, \theta_{29}) = \frac{1}{m} \sum_{i=1}^{m} \varepsilon_i^2, \qquad m = 100{,}000.} \tag{2}$$

Problem 1(e): Evaluate the learned hedge on test paths

Task: Evaluate the trained networks on the 10,000 independent test paths.

$$\mathrm{P\&L}_i = p + \sum_{j=0}^{29} H_j(s_{t_j}^{(i)})(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}) - (s_T^{(i)} - K)^+.$$

1. Compute one terminal P&L per test path.
2. Plot a histogram of these 10,000 P&L values.
3. Print their empirical mean and standard deviation.

Problem 1(f): Derive and test the analytical delta hedge

(i) Derive the benchmark: Differentiate the call-price formula (3) with respect to s:

$$\boxed{H_t^{\mathrm{BS}}(s) = \frac{\partial C(t,s)}{\partial s}.}$$

(ii) Evaluate it on the same test paths:

1. At t_j, hold $H_{t_j}^{\mathrm{BS}}(S_{t_j})$ shares until t_{j+1}.
2. Compute terminal P&L with the same p, payoff, and 30 trading intervals as in (e).
3. Plot the histogram and print the empirical mean and standard deviation.
4. Compare the analytical hedge with the learned hedge from part (e).

Problem 1(g): Compare learned holdings with analytical delta

Task: Compare the two hedge functions at four trading dates:

$$j \in \{0, 10, 20, 29\}, \qquad t_j = jT/N, \qquad s \in [0.5, 1.5].$$

1. Make a grid of stock prices over $[0.5, 1.5]$.
2. At each selected j, evaluate both functions on that same grid:

$$s \longmapsto H_j(s), \qquad s \longmapsto H_{t_j}^{\mathrm{BS}}(s).$$

3. Overlay the two curves; label the trading date and both strategies.
4. Answer: at which times are the functions most similar, and why?

Problem 2: Learn a hedge in the Heston model

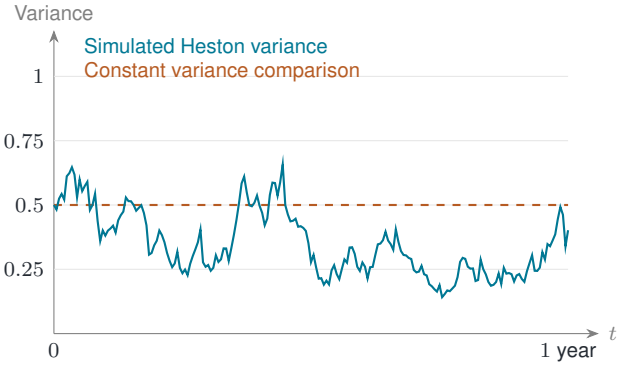
The setting: Sell the same European call, now on a stock with stochastic volatility. Some risk cannot be eliminated by trading the stock and cash alone.

What you will build

- Simulated Heston stock and variance paths.
- A hedge network using both price and variance at each trading date.
- Strategies trained to minimize AVaR for $\alpha = 0.5$ and $\alpha = 0.9$.

Goal: learn how to hedge and price the residual risk when perfect replication is generally unavailable.

Stochastic Volatility



Black–Scholes

$$\text{Variance rate} = \sigma_{\text{BS}}^2.$$

Heston

$$\text{Variance rate} = V_t, \qquad \text{volatility} = \sqrt{V_t}.$$

The stock's fluctuation size changes as V_t changes.

Mean reversion: variance fluctuates around a long-run level rather than staying fixed.

Heston allows persistent volatility changes and correlation with stock returns.

Heston model

Under the project's risk-neutral measure $\mathbb{Q}$, with $r = 0$:

$$\boxed{\begin{aligned} dS_t &= \sqrt{V_t}\, S_t\, dW_t, & S_0 &= s_0, \\ dV_t &= a(b - V_t)\, dt + \sigma\sqrt{V_t}\, dW'_t, & V_0 &= v_0. \end{aligned}} \tag{4}$$

The variance dynamics

- b: long-run variance level.
- a: speed of mean reversion.
- σ: size of random variance shocks.

If $V_t < b$, its drift is positive; if $V_t > b$, its drift is negative.

Correlated Brownian shocks

$$d\langle W, W' \rangle_t = \rho\, dt.$$

Negative ρ associates downward stock shocks with upward variance shocks.

Notation change: σ is no longer the stock's constant volatility.
The stock's instantaneous volatility is $\sqrt{V_t}$.

Residual Hedging Loss

For a strategy H, define the trading gain and the unfinanced terminal loss:

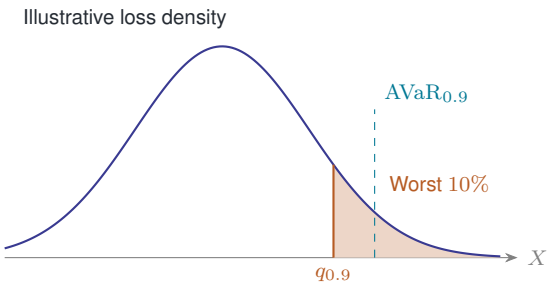
$$G(H) = \int_0^T H_u \, dS_u, \qquad \boxed{X(H) = g(S_T) - G(H).}$$

- $g(S_T)$ is the payment owed to the option buyer.
- Trading gains reduce $X(H)$; trading losses increase it.
- After receiving premium p, the seller's P&L is $p - X(H)$.

The new decision

Exact replication is generally unavailable. Choose a strategy that reduces a specified risk measure $R(X(H))$, and charge a premium that compensates for that remaining risk.

AVaR



A quantile locates the tail:

$$q_\alpha(X) = \inf\{x : \Pr(X \le x) \ge \alpha\}.$$

For a continuous loss distribution, approximately $1 - \alpha$ of losses exceed this threshold.

AVaR averages the tail:

$$\boxed{\mathrm{AVaR}_\alpha(X) = \frac{1}{1-\alpha}\int_\alpha^1 q_u(X)\,du.}$$

It accounts for how large the worst losses are, not just where they begin.

$\alpha = 0.5$: average the worst 50%. $\alpha = 0.9$: average the worst 10%.

Calculating AVaR

Suppose the sorted losses are measured in illustrative monetary units:

$$\underbrace{0,\ 1,\ 2,\ 3,\ 4}_{\text{lower half}},\qquad \underbrace{5,\ 6,\ 7,\ 8,\ 14}_{\text{worst half}}.$$

Level $\alpha = 0.5$

$$(1-\alpha)\times 10 = 5 \text{ losses},$$

$$\boxed{\mathrm{AVaR}_{0.5} = \frac{5+6+7+8+14}{5} = 8.}$$

Level $\alpha = 0.9$

$$(1-\alpha)\times 10 = 1 \text{ loss},$$

$$\boxed{\mathrm{AVaR}_{0.9} = 14.}$$

Instead of sorting the losses at every step, introduce a scalar threshold w:

$$\boxed{\mathrm{AVaR}_\alpha(X) = \inf_{w\in\mathbb{R}} \left\{ w + \frac{1}{1-\alpha}\mathbb{E}[(X-w)^+] \right\}.}$$

- w: a candidate cutoff for the worst losses.
- $(X-w)^+ = \max(X-w, 0)$: loss above that cutoff, a ReLU operation.
- $1/(1-\alpha)$: rescales the average excess to the tail fraction.

The Heston optimization

Choose the strategy with the smallest AVaR of the loss:

$$\boxed{p = \inf_H \mathrm{AVaR}_\alpha\Big(g(S_T) - \int_0^T H_u \, dS_u\Big).} \tag{5}$$

On the time grid, define each simulated path's loss using current price and variance:

$$X_i(\theta) = g(s_T^{(i)}) - \sum_{j=0}^{N-1} H_j(s_{t_j}^{(i)}, v_{t_j}^{(i)}; \theta_j)(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}).$$

Replace the expectation by an average over m paths and optimize jointly:

$$\boxed{\min_{\theta,w} \widehat{\mathcal{L}}_\alpha(\theta, w) = \min_{\theta,w} \left\{ w + \frac{1}{m(1-\alpha)} \sum_{i=1}^{m} (X_i(\theta) - w)^+ \right\}.} \tag{6}$$

- Learn all hedge-network parameters θ and one shared scalar w.
- Estimate p by evaluating the trained objective on independent test paths.
- Here p is the resulting risk-based premium; it is not a fixed Black–Scholes input.

Problem 2(a): Simulate Heston stock and variance paths

Task: Generate 100,000 training paths and 10,000 independent test paths of (S, V).

$$N = 30, \quad T = \tfrac{1}{4}, \quad s_0 = 1, \quad v_0 = b = 0.5, \quad a = 4, \quad \sigma = 1, \quad \rho = -0.3, \quad \Delta t = T/N.$$

(i) Simulate variance exactly on the grid:

$$c = \frac{\sigma^2}{4a}(1 - e^{-a\Delta t}), \qquad C_j \mid V_{t_j} \sim \chi'^2_\nu(\lambda_j), \qquad V_{t_{j+1}} = cC_j,$$

$$\nu = \frac{4ab}{\sigma^2}, \qquad \lambda_j = e^{-a\Delta t}\frac{V_{t_j}}{c}.$$

Use a noncentral chi-square sampler, such as `scipy.stats.ncx2`.

(ii) Simulate stock prices with the supplied simplified Broadie–Kaya update:

$$\boxed{S_{t_{j+1}} = S_{t_j} \exp\left[\frac{\rho}{\sigma}(V_{t_{j+1}} - V_{t_j} - ab\Delta t) + \left(\frac{a\rho}{\sigma} - \frac{1}{2}\right) V_{t_j}\Delta t + \sqrt{(1-\rho^2)V_{t_j}\Delta t}\, Z_{j+1}\right].}$$

Draw fresh $Z_{j+1} \sim \mathcal{N}(0, 1)$, independent of the variance-path simulation.

Problem 2(b): Use stock price and variance as network inputs

Task: Implement $N = 30$ neural networks, one for each trading interval:

$$\boxed{H_j : \mathbb{R}^2 \to \mathbb{R}, \qquad (S_{t_j}, V_{t_j}) \longmapsto H_j(S_{t_j}, V_{t_j}; \theta_j).}$$

- **Inputs:** the current stock price and current variance.
- **Output:** the number of stock shares held until t_{j+1}.

What changes from Problem 1?

The input now has two components. The output remains one stock position.
Observing variance helps the trading decision; it does not make variance a traded asset.

Problem 2(c): Train separate hedges for two AVaR levels

Task: Hedge the same call, $g(S_T) = (S_T - 1)^+$, with $T = 1/4$, using separate models for $\alpha = 0.5$ and $\alpha = 0.9$.

For each training path, compute the loss:

$$X_i(\theta) = (s_T^{(i)} - 1)^+ - \sum_{j=0}^{29} H_j(s_{t_j}^{(i)}, v_{t_j}^{(i)}; \theta_j)(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}).$$

Jointly optimize all network parameters and one trainable scalar threshold w:

$$\boxed{\mathcal{L}_\alpha(\theta, w) = w + \frac{1}{m(1-\alpha)} \sum_{i=1}^{m} \max\{X_i(\theta) - w, 0\}.} \tag{6}$$

- Each α has its own trained networks and its own w.
- Include w in the optimizer; it is shared across all paths, not predicted per path.

Problem 2(d): Evaluate and compare the risk-based prices

Task: Evaluate equation (6) on the 10,000 test paths for each trained model.

With trained parameters $(\widehat{\theta}_\alpha, \widehat{w}_\alpha)$, compute

$$\boxed{\widehat{p}_\alpha = \widehat{w}_\alpha + \frac{1}{n(1-\alpha)} \sum_{i=1}^{n} \left(X_i^{\text{test}}(\widehat{\theta}_\alpha) - \widehat{w}_\alpha \right)^+, \qquad n = 10{,}000.}$$

1. Freeze each strategy and its threshold; use the same test paths for both models.
2. Report $\widehat{p}_{0.5}$ and $\widehat{p}_{0.9}$.
3. Answer: which price is higher, and why?

Problem 2(e): Compare the final hedging P&L distributions

Task: Compute P&L on the test set for both AVaR levels, using each model's own price:

$$\mathrm{P\&L}_i^{(\alpha)} = \widehat{p}_\alpha + \sum_{j=0}^{29} H_j^{(\alpha)}(s_{t_j}^{(i)}, v_{t_j}^{(i)})(s_{t_{j+1}}^{(i)} - s_{t_j}^{(i)}) - (s_T^{(i)} - 1)^+.$$

1. Use the corresponding strategy and price for $\alpha = 0.5$ and $\alpha = 0.9$.
2. Plot both P&L histograms with comparable axes and binning.
3. Report the empirical mean and standard deviation for each distribution.